If $(x+y)$ and $(p+q)$ are two binomials, we can find their product as given below,

(i) Horizontal distributive approach

$$(x+y)(p+q) = x(p+q) + y(p+q)$$

$$= xp + xq + yp + yq$$

(ii) Vertical distributive approach

$$\begin{array}{r} x + y \\ p + q \\ \hline xq + yq \\ xp + yp \\ \hline = xp + xq + yp + yq \end{array}$$

(iii) Grid approach

$\times$	x	y
p	xp	yp
q	xq	yq

$= xp + xq + yp + yq$

(iv) FOIL approach

We can do it directly like, $(x+y)(p+q) = \underset{\text{Firsts}}{x(p)} + \underset{\text{Outer}}{x(q)} + \underset{\text{Inner}}{y(p)} + \underset{\text{Last}}{y(q)}$

So, the above area of the rectangle $= (l-5)\times(b+3)$

(By horizontal distributive approach) $= l(b+3) - 5(b+3)$

$A = (lb + 3l - 5b - 15)\ sq.u$

Aliter

By grid approach

$\times$	l	-5
b	lb	$-5b$
3	$3l$	-15

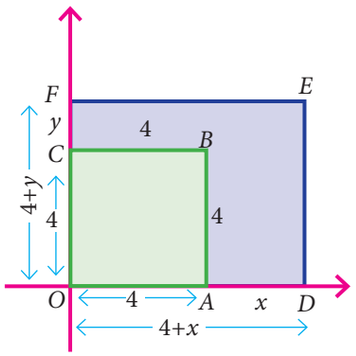
$= (lb + 3l - 5b - 15)$

Let us consider one more example. Consider the given figure , In the square $OABC$,

$OA = 4$ *units* ; $OC = 4$ *units*

The area of the square $OABC = 4 \times 4$

$A = 16$ *sq.units*



If the sides of the square are increased by 'x' *units* and 'y' *units* respectively,then we get the rectangle $ODEF$ whose sides are $OD=(4+x)$ *units* and $OF=(4+y)$ *units*.

Now, the area of the rectangle $ODEF = (4+x)(4+y)$

(by FOIL approach) $A = 16 + 4y + 4x + xy$ *sq. units.*

(all are unlike terms andso, we can't add)

Example 3.5

Multiply $(2x+5y)$ and $(3x-4y)$

Solution:

By horizontal distributive approach,

$$(2x+5y)(3x-4y) = 2x(3x-4y) + 5y(3x-4y)$$

$$= 6x^2 - 8xy + 15xy - 20y^2$$

$$= 6x^2 + 7xy - 20y^2 \text{ (simplify the like terms)}$$

Aliter

By grid approach

$\times$	$2x$	$5y$
$3x$	$6x^2$	$15xy$
$-4y$	$-8xy$	$-20y^2$

$= 6x^2 - 8xy + 15xy - 20y^2$

$= 6x^2 + 7xy - 20y^2$

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